

EE798I: Assignment 4

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Answer 3: Diffraction grating

The design is governed by the grating equation for reflection, which relates the incident angle (θ_i), diffracted angle (θ_d), wavelength (λ), grating period (Λ), and diffraction order (m):

$$n_i \sin(\theta_d) = n_i \sin(\theta_i) + \frac{m\lambda}{\Lambda}$$

- **Incident Medium:** Air, so $n_i \approx 1$.
- **Wavelength (λ):** We choose a standard telecom wavelength, $\lambda = 1550$.
- **Incident Angle (θ_i):** The design specifies $\theta_i = \pi/4$ or 45° .
- **Diffracted Angle (θ_d):** For retro-reflection, we need $\theta_d = -\pi/4$ or -45° .
- **Diffraction Order (m):** This specific reflection corresponds to the $m = -1$ order.

We plug these values into the grating equation to solve for the unknown period, Λ :

$$\begin{aligned} 1 \cdot \sin(-45^\circ) &= 1 \cdot \sin(45^\circ) + \frac{(-1) \cdot 1550}{\Lambda} \\ -\sin(45^\circ) &= \sin(45^\circ) - \frac{1550}{\Lambda} \\ \frac{1550}{\Lambda} &= 2 \cdot \sin(45^\circ) \\ \frac{1550}{\Lambda} &= 2 \cdot \left(\frac{\sqrt{2}}{2} \right) \\ \frac{1550}{\Lambda} &= \sqrt{2} \\ \Lambda &= \frac{1550}{\sqrt{2}} \approx 1096 \end{aligned}$$

The provided COMSOL plots (Figure 1 and 2) successfully confirm that the physical phenomenon of diffraction is occurring as intended.

- **Figure 1 (Plot of E_z):** The checkerboard-like pattern at the top is the result of the incident wave ($\theta_i = \pi/4$) interfering with the various reflected diffracted orders.
- **Figure 2 (Plot of normE):** The high-intensity (red) and low-intensity (blue) spots are regions of constructive and destructive interference, respectively. This strong, angled interference pattern confirms that a significant portion of the incident wave is being reflected at an angle other than the specular ($m = 0$) angle.

These plots confirm that the $m = -1$ diffracted order exists.

The grating equation for reflection is:

$$\sin(\theta_d) = \sin(\theta_i) + \frac{m\lambda}{\Lambda}$$

Our design targets the $m = -1$ order, where $\theta_d = -\pi/4$.

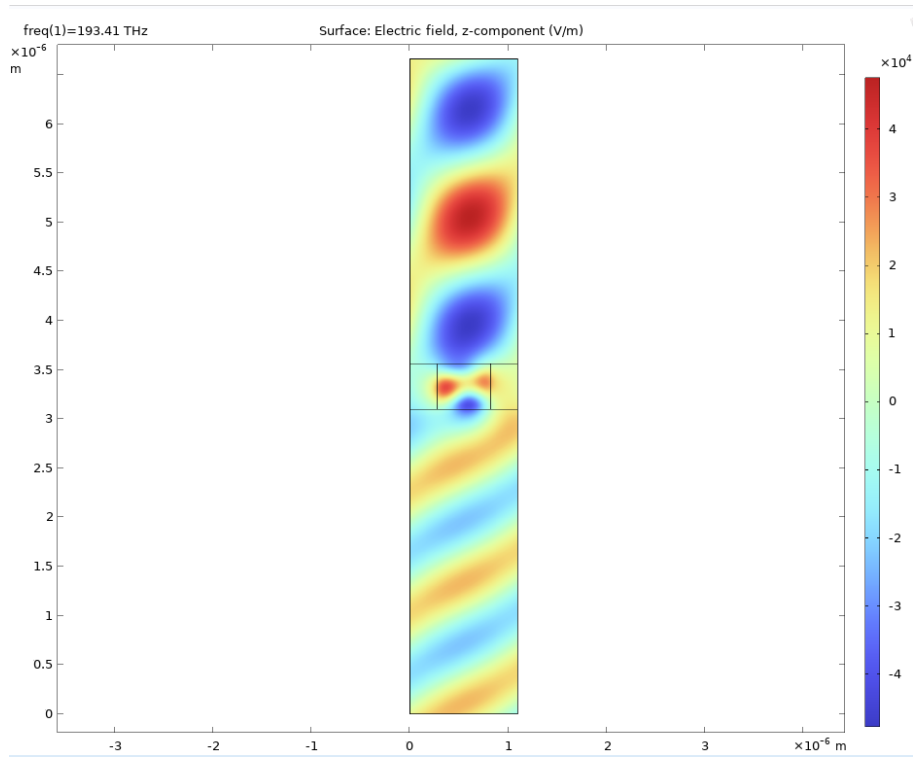


Figure 1: Plot of E_z

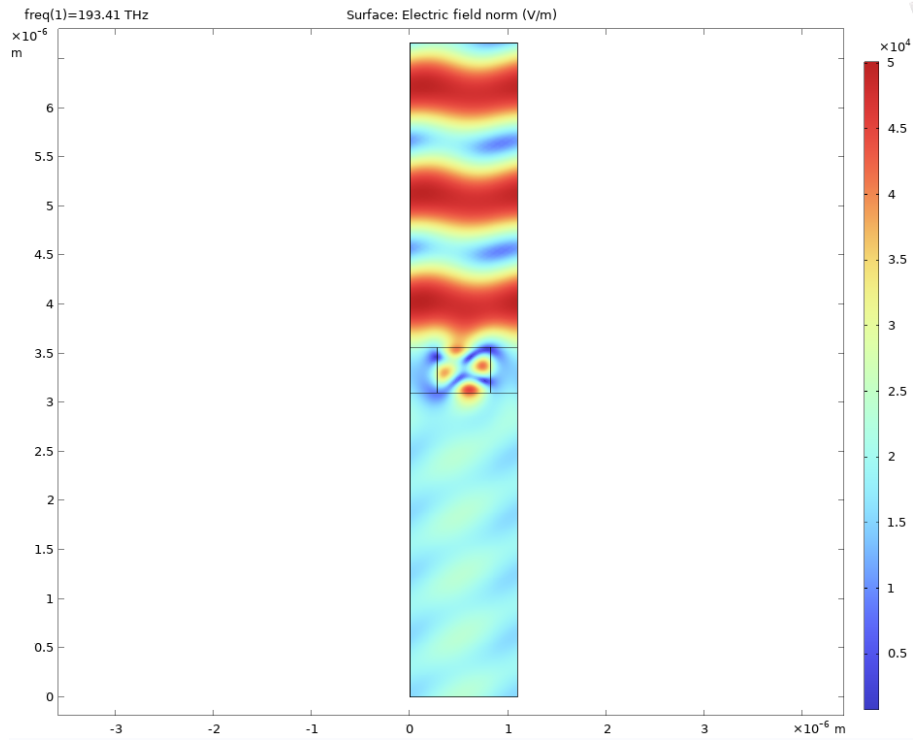


Figure 2: Plot of normE

Bonus Question: Guided modes of a waveguide with finite cross-section

The material refractive indices at this wavelength are taken as:

- Core (Si): $n_{\text{core}} = 3.47$
- Cladding (SiO): $n_{\text{clad}} = 1.44$

The modes are identified by their field distribution (number of horizontal lobes) and their effective mode index (n_{eff}):

- **Fundamental Mode ($\text{TE}_{0,0}$):** This is shown in **Figure 4(f)**. It has the highest effective index, $n_{\text{eff}} = 2.8338$, and a single lobe, indicating it is the fundamental mode in both x and y directions.
- **Higher-Order Modes:** The other plots show higher-order horizontal modes, which have $m > 0$ lobes.
 - **Figure 4(e)** ($n_{\text{eff}} = 2.8212$): $\text{TE}_{1,0}$ mode (2 horizontal lobes).
 - **Figure 4(d)** ($n_{\text{eff}} = 2.7995$): $\text{TE}_{2,0}$ mode (3 horizontal lobes).
 - **Figure 4(c)** ($n_{\text{eff}} = 2.7689$): $\text{TE}_{3,0}$ mode (4 horizontal lobes).
 - **Figure 4(b)** ($n_{\text{eff}} = 2.7291$): $\text{TE}_{4,0}$ mode (5 horizontal lobes).
 - **Figure 4(a)** ($n_{\text{eff}} = 2.6794$): $\text{TE}_{4,0}$ mode (6 horizontal lobes).

All guided modes correctly have $n_{\text{clad}} < n_{\text{eff}} < n_{\text{core}}$.

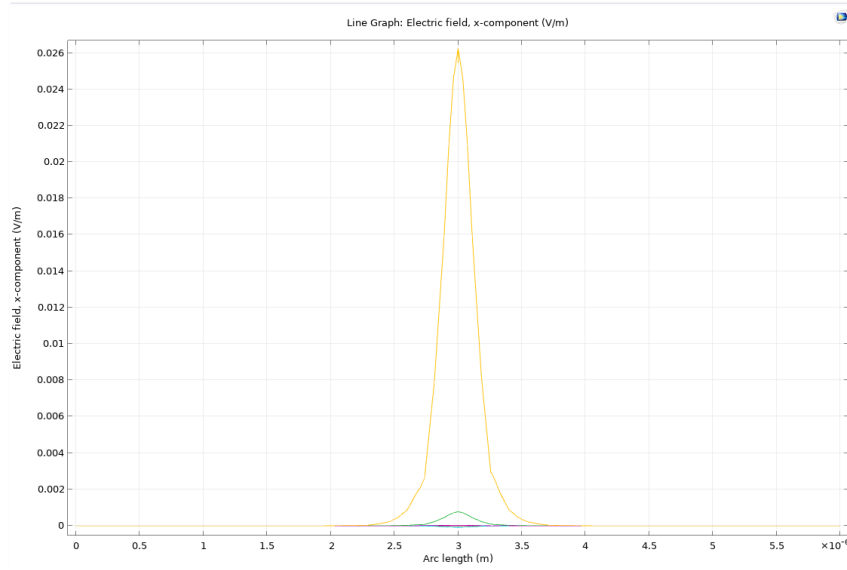


Figure 3: 2D CutLine Plot of E_x

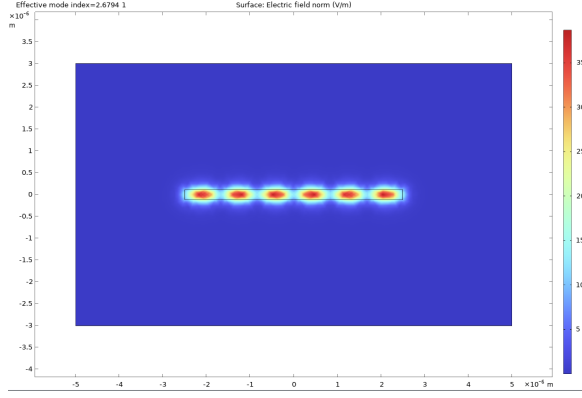
We compare the fundamental $\text{TE}_{0,0}$ mode (**Figure 4(f)**).

For a 1D slab, the dominant field component (E_x) for the fundamental TE mode is:

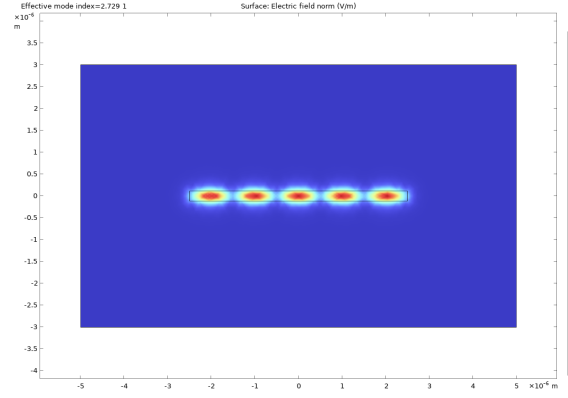
- **Inside Core** ($|y| \leq H/2$): $E_x(y) \propto \cos(\kappa_y y)$
- **In Cladding** ($|y| > H/2$): $E_x(y) \propto \exp(-\gamma_y |y|)$

The numerical field distribution is analyzed using the 1D cut-line **Figure 3**. This plot shows the E_x field component along a vertical line through the center of the waveguide.

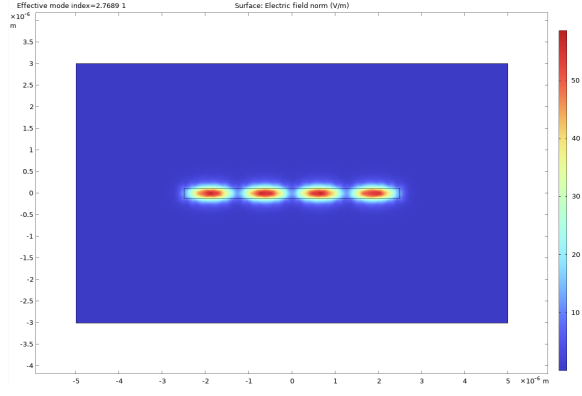
1. **Vertical Profile:** The yellow curve (representing the fundamental mode) in the cut-line plot shows the exact shape predicted by the analytical solution: a symmetric, cosine-like peak centered at the core, which rapidly decays exponentially into the cladding.
2. **Horizontal Profile:** The 2D surface (**Figure 4(f)**) shows that the field is almost perfectly constant along the x -direction, only decaying near the edges. This confirms the slab approximation ($d/dx \approx 0$).



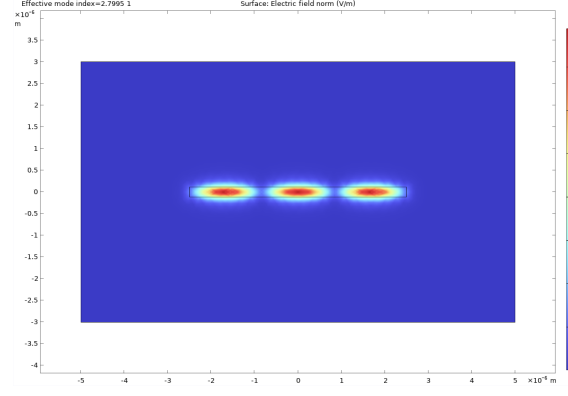
(a) Mode with $n_{\text{eff}} = 2.6794$



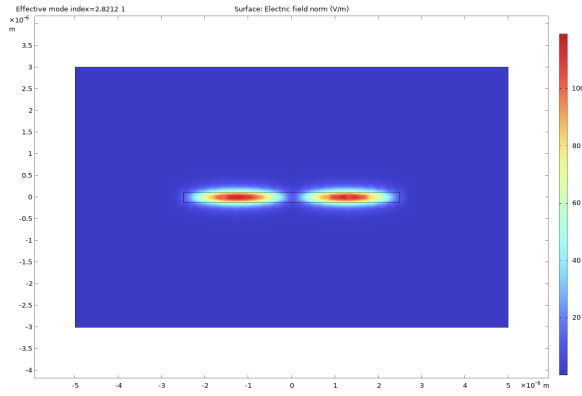
(b) Mode with $n_{\text{eff}} = 2.7291$



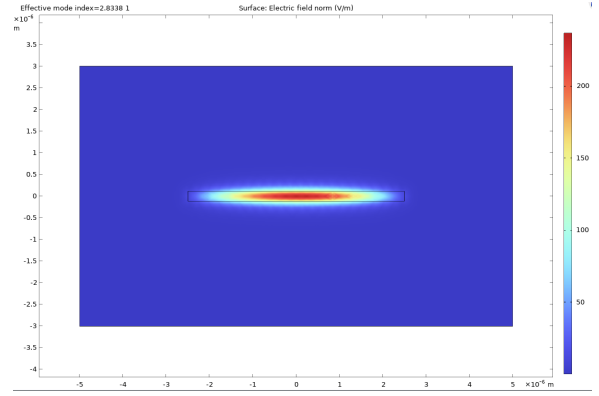
(c) Mode with $n_{\text{eff}} = 2.7689$



(d) Mode with $n_{\text{eff}} = 2.7995$



(e) Mode with $n_{\text{eff}} = 2.8212$



(f) Fundamental Mode, $n_{\text{eff}} = 2.8338$

Figure 4: Surface plots of the electric field norm (normE) for the six guided modes with the highest effective indices.